Amazon Marketplace Product Intelligence Platform

YSD Training Program-Batch05

1. Background

Our team manually checks Amazon listings one by one. We don't have one central place to see product prices, how they look, and customer feedback. We need a system that automatically collects this information from live Amazon listings and puts it in a single application where non-technical staff can explore and analyze the data.

2. Objective

The project has three parts of deliverables that work together. First, you need to build your own data collection pipeline to gather fresh, verifiable Amazon product and review data; this can't be a pre-existing dataset. Second, using that data, you'll build a single application with four analytical features that extract insights from the products and reviews.

Main Goal

Part A: Data

- Collect real, up-to-date Amazon data for a given list of products
- Include: product info + customer reviews
- Must be scraped/collected by you (your own pipeline), not downloaded from an existing dataset

Part B: Analysis & Model Development

- Prepare notebooks for each analytical feature model development stage
- Preprocess the raw dataset
- Analyze the data
- Develop models.

Part C: Application

- Build one single application (use Streamlit)
- It should run on top of your collected data
- It must have 4 features (explained below)

3. Scope of Work

3.1. Data collection

- **Input:** You will receive an Excel item list (search keyword). Work only from your assigned list;
  - **Coverage:** For each assigned search keyword, search the site and collect data from all unique product listings displayed on the first page of the search results, where available. For each product, collect all individual reviews visible on the first page of reviews.
  - **Automation:** Handle loading content using code. Do not copy-paste manually or use purchased or existing datasets.
  - **Logging:** Keep a log of every request's result (success, blocked, or empty), with the time it happened.

- **Output:** A single Excel or CSV sheet with one row per product. Each row should contain the product information, with all collected reviews for that product stored in a single reviews field as an array of JSON objects.

3.2. Data preparation and analysis

Data preparation and analysis will be performed separately for each analytical feature (A–D). Each feature will follow its own pipeline. Prepare notebooks for each case:

- **Preprocessing:** Clean and prepare the relevant data, including handling duplicates, missing values, invalid data, and required transformations.
- **EDA & Analysis:** Explore the relevant data, identify patterns, relationships, data quality issues, and useful attributes.
- **Interpretation:** Document key findings and how they influence the selected approach.
- **Model Development & Comparison:** Develop multiple suitable models or approaches, evaluate and compare their performance, and identify the strongest candidates.
- **Fine-tuning & Final Selection:** Fine-tune the selected candidates and choose the final model or approach based on the defined evaluation criteria.

3.3. Analytical features

All four features will follow the pipeline defined in Section 3.2, from preprocessing and EDA through analysis, model development, comparison, fine-tuning, and final selection.

- **A - Similar product recommendation:** Take a product or search term. Return a list of similar products ranked by relevance, with a short reason for each. It must work for all products and give the same results for the same input.
- **B - Review sentiment:** Sort each review as positive, neutral, or negative. Create summaries by product and by category. Reviews come without labels; you decide how to get and check labels (document this). Test your accuracy against reviews you checked manually.
- **C - Thumbnail grouping:** Group product images into visual groups without using the site's category labels. Show sample images and group size for each group. Say how many groups you made and why. You can use category labels later to check if groups make sense.
- **D - Price tier classification:** Classify each product into a price tier (e.g. budget, mid-range, premium) using product attributes such as title, description, brand, and review count etc. not the price itself.

3.4. Application

A single application, that:

- Show a list of products.
- Show details about each product.
- Let users run each of the four features.

4. Constraints

- Collect only from publicly accessible pages. Do not bypass logins, CAPTCHAs, or other access controls, and keep request rates modest.
- Do not collect personal data beyond the reviewer's names publicly displayed on the listing.
- All work must sit in a version-controlled repository with regular, incremental git commits.

5. Deliverables

- Collected dataset (products and reviews), and the request/response log.
- Collection pipeline source code, runnable end to end.
- Cleaned dataset and the cleaning pipeline.
- Analysis & model development notebooks.
- The four features: saved models/outputs, and a short evaluation note.
- Application source code and setup instructions.
- A 15-minute walkthrough of the delivered system, with live questions.

6. Milestones

Milestone	Due	Submitted for review
M1 - Dataset	End of day 2	Collected dataset, collection pipeline, raw evidence, log
M2 – Analysis & Development	End of day 5	Cleaned dataset, cleaning pipeline, analysis & model development notebook
M3 - Application	End of day 7	Integrated application, setup instructions, final walkthrough

7. Acceptance Criteria

- Check 5-10% of product URLs to make sure the information matches the actual website. Video links should work.
- The application works using just the instructions you were given, with all four features available.
- Your crawling/scrapping log is up to date and matches the work you did.
- You can explain any part of what you built and change any part if asked.

8. Questions

Ask questions if something is unclear. Don't guess what it means. Write down all questions and answers in a document called "Project Questions & Clarifications Log". This document is the official record for the project. If something is not written in this document, it does not count.